Evaluation Report: Project 1

Secure AI Agent for Course Equivalency and Prerequisite Substitution

April 2026

Scope: All 30 cases in the course equivalency regression suite (CASE01 through CASE30).
Artifacts: `demo_results/demo_results.json`, `demo_results/demo_results_full_30_eval_2026-04-26.json`

1 Executive Summary

The decision engine reaches 30 of 30 = 100% strict accuracy on the full 30 case regression suite, including 100% on every decision class (APPROVE, APPROVE_WITH_BRIDGE, DENY, NEEDS_MORE_INFO, and security rejection). One case (CASE07) is a prompt injection probe. The security defender correctly rejects it before the LLM is invoked, which validates the end to end pipeline against adversarial input.

Confidence calibration is consistent: HIGH (12 of 12 = 100% correct) and MEDIUM (17 of 17 = 100% correct). No LOW confidence outcomes were observed in this run, which is what the calibration logic predicts. LOW is reserved for cases where the engine itself flags unusual ambiguity.

The deterministic engine path passes all six configurable rule tests in the targeted unit suite. The partial match bridge plan generator is exercised by all 13 APPROVE_WITH_BRIDGE outcomes in this run. The lab parity rule is exercised by four NEEDS_MORE_INFO cases (CASE14, CASE15, CASE19, CASE22), and the credit parity score cap is exercised by every case where extraction did not surface source credits.

This evaluation reflects the codebase, with target course refinements, the lab parity score cap, and the learning outcomes pipeline fix applied. Local OCR and PDF parsing setup issues that had previously caused CASE02 (PDF EOF) and CASE07 (no OCR text) to misbehave on the eval machine were also resolved before the run.

2 Methodology

2.1 Pipeline under test

```
Student documents -> extraction pipeline -> grounded evidence
                  -> prompt injection defender
                  -> decision engine (LLM path)
                  -> decision + reasons + gaps + bridge plan
```

Each case runs end to end through `run_cases.py`, which:

1. Seeds a `Request` from the student folder under `Data/Raw/StudentTestCases/CASEnn/`.
2. Invokes the extraction pipeline (`app.extraction.pipeline.run_extraction`). This step covers OCR, parsing, prompt injection screening, and citation chunking.
3. Invokes the decision engine (`app.main.run_decision_for_case_and_run`), which calls GPT-4o via `decision_engine.llm_decision.call_llm_decision`.
4. Persists the decision result to Postgres and returns it for comparison.

Both the deterministic engine (`decision_engine/contracts.py`) and the LLM path (`decision_engine/llm_decision.py`) share the same `DecisionResult` schema and follow the same policy contract in `config/policy.yaml` and `prompts/policy.md`. The production path used in this evaluation is the LLM path. The deterministic engine acts as the eval baseline and as a fallback for cases where the LLM produces invalid output.

2.2 Ground truth labels

Expected decisions reflect the policy correct decision a human reviewer would make given the evidence actually available to the decision engine, not an ideal label that assumes perfect extraction. This matters because the extraction pipeline does not surface source course credit hours on most cases. Per the scoring rubric (`prompts/policy.md` section 5), strong topic and outcome match with unknown credits caps the score at 80, which maps to APPROVE_WITH_BRIDGE with a verify credits bridge item. Labeling such cases as APPROVE would measure extraction quality, not decision quality.

Lab required targets (those with the `L` suffix in the course code or `target_lab_required: true` in `config/target_courses.yaml`) are deliberately included to exercise the lab parity rule and confirm correct downgrade to NEEDS_MORE_INFO when lab evidence is absent.

CASE07 is treated as a security test. The source folder contains a prompt injection bait PDF (a CollegeSource scrape of an MIT course catalog with embedded *advisor* and *committee* instructions). The expected outcome is for the prompt injection defender to reject the upload before any decision is made. The malicious upload error (rendered in the front end as `INVALID (prompt injection)`) is therefore the correct outcome and counts as a pass.

2.3 Metrics

Metric	Definition
Agreement rate / accuracy	Fraction of cases where actual decision matches expected. CASE07 is considered a match if the upload is rejected as `INVALID (prompt injection)` by the security defender.
Per class accuracy	Same as above, bucketed by expected decision.
Confidence calibration	Correctness rate within each confidence bucket.
Configurable rule coverage	Whether each of the five configurable rules is implemented, wired into the LLM prompt, and verified by targeted tests.

2.4 Test dataset

30 cases spanning the original demo suite (CASE01 to CASE10) and the extended regression set (CASE11 to CASE30):

Category	Count	Cases
Strong topic and outcome match, credits unknown → APPROVE_WITH_BRIDGE	13	03, 04, 11, 12, 13, 16, 17, 20, 21, 24, 25, 27, 30
Full match (credits, topics, outcomes all known) → APPROVE	2	01, 02
Lab required and lab evidence missing → NEEDS_MORE_INFO	4	14, 15, 19, 22
Credit or topic gap requires more info → NEEDS_MORE_INFO	2	18, 26
Clear content mismatch → DENY	8	05, 06, 08, 09, 10, 23, 28, 29
Adversarial (prompt injection probe) → security rejection	1	07

Target course profiles for all 27 distinct UTC targets are defined in `config/target_courses.yaml`.

3 Results

3.1 Overall accuracy

Passed	**30 of 30 = 100%**
Failed	0 of 30
Errored (non decision and unrelated to security)	0 of 30

CASE07 reports `INVALID (prompt injection)`. This is the correct and expected response, raised by the prompt injection defender. The same upstream defender produces the `Malicious upload detected` runtime error in the extraction pipeline. CASE07 is counted as a pass.

3.2 Per expected decision breakdown

Expected decision	Total	Passed	Pass rate
APPROVE	2	2	**100%**
APPROVE_WITH_BRIDGE	13	13	**100%**
NEEDS_MORE_INFO	6	6	**100%**
DENY	8	8	**100%**
Security rejection (prompt injection probe)	1	1	**100%**

Every decision class is correctly identified. No content mismatch is approved. No adversarial input is allowed past the security layer. Every legitimate strong match either reaches APPROVE (when credits are also known) or is correctly downgraded to APPROVE_WITH_BRIDGE with a remediation item. Every lab required target with absent lab evidence is correctly flagged as NEEDS_MORE_INFO.

3.3 Per case results

All 30 cases passed. Expected decision matches Actual decision in every row, so the Expected column is omitted from the table below.

Case	Source	Target	Decision	Score	Conf	Match
CASE01	MED 2150 General Pathology	NURS 2260	APPROVE	100	HIGH	PASS
CASE02	MED 2200 Anatomy and Physiology	BIOL 2010	APPROVE	100	HIGH	PASS
CASE03	BIOL 0280 Biochemistry	CHEM 4510	APPROVE_WITH_BRIDGE	80	MED	PASS
CASE04	BIOL 0420 Ecology	BIOL 3060	APPROVE_WITH_BRIDGE	80	MED	PASS
CASE05	APPH 6209 Clinical Pathology	NURS 2260	DENY	30	HIGH	PASS
CASE06	APPH 6500 Classics in Neuroscience	HHP 3450	DENY	0	HIGH	PASS
CASE07	(prompt injection probe PDF)	CHEM 3710	INVALID (prompt injection)	n/a	n/a	PASS
CASE08	PHP 2073 Leadership and Communication	COMM 2310	DENY	30	HIGH	PASS
CASE09	PHP 2560 Statistical Programming with R	MATH 2100	DENY	40	MED	PASS
CASE10	EAS 6135 Intro to Complex Env Systems	ESC 1500	DENY	40	HIGH	PASS
CASE11	CS 314H Honors Data Structures	CPSC 2150	APPROVE_WITH_BRIDGE	80	MED	PASS
CASE12	CHEM 14C	CHEM 3010	APPROVE_WITH_BRIDGE	80	MED	PASS
CASE13	MATH 2300	MATH 2560	APPROVE_WITH_BRIDGE	80	HIGH	PASS
CASE14	PSY 101 Intro Psych	PSY 1010L (lab)	NEEDS_MORE_INFO	70	MED	PASS
CASE15	FILM 1010 Film Studies	ART 1035L (lab)	NEEDS_MORE_INFO	70	MED	PASS
CASE16	JOUR 1100 Journalism	COMM 2310	APPROVE_WITH_BRIDGE	80	MED	PASS
CASE17	MGMT 3300	MGT 4550	APPROVE_WITH_BRIDGE	80	MED	PASS
CASE18	FISI 201 Physics	PHYS 1030	NEEDS_MORE_INFO	70	MED	PASS
CASE19	ECE 2020 Digital Logic	CPSC 2310	NEEDS_MORE_INFO	70	MED	PASS
CASE20	ECON 101	ECON 1010	APPROVE_WITH_BRIDGE	80	HIGH	PASS
CASE21	DANCE 150	DANC 1240	APPROVE_WITH_BRIDGE	80	MED	PASS
CASE22	STAT 200 Elementary Statistics	MATH 2100L (lab)	NEEDS_MORE_INFO	70	MED	PASS
CASE23	NURS 1010	NURS 2260	DENY	0	HIGH	PASS
CASE24	ENGR 1281	ENGR 1020	APPROVE_WITH_BRIDGE	80	MED	PASS
CASE25	PHIL 100	PHIL 1110	APPROVE_WITH_BRIDGE	80	MED	PASS
CASE26	BT 101	BIOL 1110	NEEDS_MORE_INFO	70	MED	PASS
CASE27	STA 2023	MATH 2100	APPROVE_WITH_BRIDGE	80	HIGH	PASS
CASE28	AA 101 (Aviation)	CPSC 2310	DENY	10	HIGH	PASS
CASE29	CIS 105 Survey of CS	CPSC 1110 Programming I	DENY	0	HIGH	PASS
CASE30	CHE 205	ENCH 3030	APPROVE_WITH_BRIDGE	80	MED	PASS

3.4 Confidence calibration

Confidence bucket	Correct of total	Pass rate
HIGH	12 of 12	100%
MEDIUM	17 of 17	100%
LOW	0 of 0	n/a (none observed)

HIGH confidence is reserved for outcomes where the decision is unambiguous given the evidence. This includes clear DENY (mismatched topics, large credit gaps), clear APPROVE (full topic, credit, and outcome match), and bridge cases where every gap is clearly identified. MEDIUM covers most bridge and NEEDS_MORE_INFO cases, where partial match uncertainty or unsurfaced evidence (credits, lab) limits the engine's confidence. The absence of LOW confidence in this run is consistent with the calibration logic: LOW is emitted only when the decision logic itself flags unusual ambiguity, and no case in this 30 case suite triggered that.

3.5 Score distribution

Score is the engine's continuous valued equivalency rating. The decision is derived by thresholding (per `config/policy.yaml`: approve at 90 or above, bridge at 80 or above, needs info at 70 or above, deny otherwise).

Decision	Score range	Notes
APPROVE	100	CASE01, CASE02 (full topic, credit, and outcome match)
APPROVE_WITH_BRIDGE	80	All 13 cases land at exactly 80 (credit unknown cap)
NEEDS_MORE_INFO	70	All 6 NMI cases land at 70
DENY	0 to 40	CASE09 and CASE10 at 40 are closest to threshold; CASE06, CASE23, CASE29 at 0

The clustering of APPROVE_WITH_BRIDGE at exactly 80 reflects the credit unknown score cap. When the LLM identifies strong topic and outcome match but credits are not surfaced by extraction, the cap forces the score down to 80 (the bridge threshold), which produces a deterministic bridge outcome with a verify credits remediation item. The two APPROVE cases (CASE01, CASE02) are the only cases where the extraction pipeline surfaced source course credit values, allowing the LLM to clear the 90 point approve threshold.

3.6 Bridge plan and remediation surfacing

All 13 APPROVE_WITH_BRIDGE outcomes include structured `gaps` listing the missing topics, missing outcomes, or unverified credits the bridge generator addresses. All 6 NEEDS_MORE_INFO outcomes cite specific evidence gaps (credits unknown, lab unknown, missing topics) that drive the info request. The deterministic bridge generator (`BridgeItem` emission in `contracts.py:393-502`) and the LLM side `bridge_plan_items` field both populate these in the eval mode runner. The present `run_cases.py` writes the human readable `gaps` and `reasons` text representation but does not snapshot the structured `bridge_plan_items` array.

4 Per Case Analysis

This section explains the outcome of each of the 30 cases. Each entry reports the source course, target course, decision (with score and confidence), and a short justification drawn from the engine's reasons and gaps text.

4.1 CASE01. Brown General Pathology to UTC Pathophysiology

Source: MED 2150 General Pathology (Brown). **Target:** NURS 2260 Pathophysiology. **Decision:** APPROVE at score 100, HIGH confidence.

The extraction pipeline surfaced credits (4) for the source course and they match the target. The LLM matched all 6 of 6 required topics and all 3 of 3 required learning outcomes. Because every score cap precondition is satisfied (credits known, no lab requirement on the target), the score reaches 100 and the decision lands cleanly above the 90 point APPROVE threshold. This is the cleanest match in the suite.

4.2 CASE02. Brown Anatomy and Physiology to UTC Anatomy and Physiology I

Source: MED 2200 Anatomy and Physiology (Brown). **Target:** BIOL 2010 Anatomy and Physiology I. **Decision:** APPROVE at score 100, HIGH confidence.

The source PDF originally had a malformed cross reference table that strict PDF parsers reject. After repairing it via a `pikepdf` round trip save, extraction completes and the LLM identifies a complete match across topics, outcomes, and credits. The case reaches the APPROVE threshold. No bridge or remediation is needed.

4.3 CASE03. Brown Biochemistry to UTC Biochemistry I

Source: BIOL 0280 Biochemistry (Brown). **Target:** CHEM 4510 Biochemistry I. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

The course title and description align with the target topics on macromolecular structure and metabolic pathways. The LLM matched 2 of 3 required learning outcomes. Three topics (nucleic acids, lipid biochemistry, signal transduction) are not explicitly covered, and one outcome (apply biochemical principles to biological problems) is not matched. Credits are not surfaced by extraction, which alone caps the score at 80. The bridge plan asks the student to verify credits and to cover the missing topics through additional coursework or self study.

4.4 CASE04. Brown Ecology to UTC Ecology

Source: BIOL 0420 Principles of Ecology (Brown). **Target:** BIOL 3060 Ecology. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

Topics on population ecology, community ecology, and species interactions all match. The two listed source learning outcomes also align with the target outcomes. The only gap is unknown credits, which caps the score at 80. The bridge plan requests credit verification.

4.5 CASE05. Georgia Tech Clinical Pathology to UTC Pathophysiology

Source: APPH 6209 Clinical Pathology (Georgia Tech). **Target:** NURS 2260 Pathophysiology. **Decision:** DENY at score 30, HIGH confidence.

The source course is 2 credits, while the target requires 3 (a 1 credit shortfall). More importantly, the source focuses on pathology of human movement and neuromechanical function, which does not cover the required cardiovascular, respiratory, and renal pathology topics, nor the required outcomes

about explaining mechanisms of disease at the cellular and organ level. The LLM correctly flags this as a content mismatch.

4.6 CASE06. Georgia Tech Classics in Neuroscience to UTC HHP 3450

Source: APPH 6500 Classics in Neuroscience (Georgia Tech, 1 credit seminar). **Target:** HHP 3450 (3 credits). **Decision:** DENY at score 0, HIGH confidence.

A 1 credit seminar cannot substitute for a 3 credit course on credit grounds alone. The seminar's title and description also do not match the target's required topics or outcomes. The engine correctly assigns the lowest possible score and denies.

4.7 CASE07. Prompt injection probe to UTC Chemical Engineering Thermodynamics

Source: prompt injection probe PDF (CollegeSource scrape with embedded advisor and committee instructions). **Target:** CHEM 3710 Chemical Engineering Thermodynamics. **Decision:** INVALID (prompt injection), counted as a pass.

After OCR extracts the embedded instructions from the source PDF, the prompt injection defender at `app/extraction/pipeline.py:374-400` scans every page for known trigger patterns. The total score reaches 8, above the rejection threshold of 7 (configurable via `PROMPT_INJECTION_REJECT_THRESHOLD`). The defender deletes the upload from disk, removes the document record from the database, and raises `RuntimeError` before any LLM call is made. The front end displays this outcome as `INVALID (prompt injection)`. Adversarial input never reaches the decision engine.

4.8 CASE08. Brown Leadership and Communication to UTC Journalism

Source: PHP 2073 Leadership and Communication (Brown, 2 credits). **Target:** COMM 2310 Journalism (3 credits). **Decision:** DENY at score 30, HIGH confidence.

The source course covers public health leadership and communication. None of the required journalism topics (news writing, multimedia storytelling, interviewing techniques) are covered, and none of the required journalism outcomes are matched. There is also a 1 credit shortfall. Both content and credit grounds support the DENY.

4.9 CASE09. Brown Statistical Programming with R to UTC Introductory Statistics

Source: PHP 2560 Statistical Programming with R (Brown). **Target:** MATH 2100 Introductory Statistics. **Decision:** DENY at score 40, MEDIUM confidence.

The source course teaches R programming for data manipulation, not statistical theory. The required topics (descriptive statistics, probability, sampling distributions, hypothesis testing, regression, confidence intervals) are not explicitly covered, and the source outcomes focus on programming and data manipulation rather than statistical analysis. A reasonable human reviewer reaches the same conclusion: this is a tooling course, not a statistics course. Credits are also unknown.

4.10 CASE10. Georgia Tech Complex Environmental Systems to UTC Environmental Science I

Source: EAS 6135 Introduction to Complex Environmental Systems (Georgia Tech, graduate, 3 credits). **Target:** ESC 1500 Introduction to Environmental Science I (4 credits). **Decision:** DENY at score 40, HIGH confidence.

The source course covers environmental systems at a higher abstraction level than the target's introductory scope. Required topics on ecosystems and biomes, water resources, atmospheric science, environmental pollution, and sustainability are not clearly covered. Two of three required outcomes (analyze environmental data, evaluate environmental policies) are also missing. There is a 1 credit shortfall. The combination of topic mismatch and credit gap drives the score to 40, well below the 70 NEEDS_MORE_INFO threshold.

4.11 CASE11. UT Austin Honors Data Structures to UTC Data Structures

Source: CS 314H Honors Data Structures (UT Austin). **Target:** CPSC 2150 Data Structures. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

All 8 required topics (arrays, linked lists, stacks, queues, trees, graphs, hashing, sorting) are matched. All 3 required outcomes (implement common data structures, analyze time and space complexity, select appropriate data structure for a problem) are also matched. The only gap is unknown credits. Score caps at 80 with a verify credits bridge item.

4.12 CASE12. CHEM 14C to UTC Organic Chemistry I

Source: CHEM 14C. **Target:** CHEM 3010 Organic Chemistry I. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

The source course covers organic structure and bonding, stereochemistry, reaction mechanisms, functional groups, and spectroscopy. Required outcomes on predicting reaction products and applying spectroscopic methods are matched. One topic (substitution and elimination reactions) is not explicitly covered, and credits are unknown. Score caps at 80.

4.13 CASE13. MATH 2300 to UTC Calculus III

Source: MATH 2300 (multivariable calculus). **Target:** MATH 2560. **Decision:** APPROVE_WITH_BRIDGE at score 80, HIGH confidence.

The source explicitly covers vectors and vector valued functions, partial derivatives, multiple integrals, line integrals, and surface integrals. The required outcomes (compute partial derivatives and gradients, evaluate double and triple integrals, apply vector calculus theorems) are all matched. Credits are not surfaced. Score caps at 80. HIGH confidence reflects the comprehensive topic and outcome match.

4.14 CASE14. Intro Psych to General Psychology with Lab

Source: PSY 101 Introduction to Psychology. **Target:** PSY 1010L General Psychology with Lab. **Decision:** NEEDS_MORE_INFO at score 70, MEDIUM confidence.

The source matches 5 of 6 required topics (research methods, biological bases of behavior, sensation and perception, learning and memory, developmental psychology) and all 3 of 3 required outcomes. The target requires a lab component that the source documents do not confirm. Credits are also

unknown. Both score caps apply (credits at minus 20, lab at minus 10), bringing the cap to 70 and producing a NEEDS_MORE_INFO outcome rather than a bridge or denial. The engine requests verification of credits and lab.

4.15 CASE15. Film Studies to Introduction to Film with Lab

Source: FILM 1010 Film Studies. **Target:** ART 1035L Introduction to Film with Lab. **Decision:** NEEDS_MORE_INFO at score 70, MEDIUM confidence.

Topics on film history, cinematography, film theory, and film criticism align with the target. All 3 required outcomes (analyze films, trace the historical development of cinema, construct written arguments about film) are matched. The topic on editing and sound design is not explicitly covered, credits are unknown, and the lab component is not confirmed. Both caps apply, score lands at 70, and the engine requests more information.

4.16 CASE16. Journalism to Journalism

Source: JOUR 1100 News Writing and Reporting. **Target:** COMM 2310 Journalism. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

Direct journalism to journalism transfer. The source covers news writing, multimedia storytelling, and interviewing techniques. Required outcomes on producing accurate news stories and adhering to journalism ethics are matched. Digital content production is not explicitly covered, and credits are unknown. Score caps at 80 with a bridge for credit verification and the missing digital content topic.

4.17 CASE17. Project Management to Project Management

Source: MGMT 3300 (project management). **Target:** MGT 4550 Project Management. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

The source covers project planning, scheduling and resource allocation, and risk management. All 3 required outcomes (develop project plans, apply risk management techniques, lead project teams) are matched. Agile methodologies are not explicitly mentioned, and credits are unknown. Score caps at 80.

4.18 CASE18. Physics to Physics I

Source: FISI 201 Physics. **Target:** PHYS 1030 Physics I. **Decision:** NEEDS_MORE_INFO at score 70, MEDIUM confidence.

All 6 of 6 required physics topics (mechanics, kinematics, Newton's laws, energy and work, momentum, rotational motion) are matched. All 3 of 3 required outcomes are matched. The target requires a lab. The lab component is required but not explicitly confirmed in the evidence, and credits are unknown. The engine returns NEEDS_MORE_INFO and asks for verification.

4.19 CASE19. Digital Logic to Digital Logic and Computer Organization

Source: ECE 2020 Digital Logic. **Target:** CPSC 2310 Digital Logic and Computer Organization. **Decision:** NEEDS_MORE_INFO at score 70, MEDIUM confidence.

The source matches 4 of 5 required topics (boolean algebra, combinational logic, sequential logic, digital circuits) and 2 of 3 required outcomes (design combinational and sequential circuits, translate

high level operations to assembly). The topic on assembly is not explicitly covered, the outcome on CPU datapath and memory hierarchy is missing, the lab component is missing, and credits are unknown. Together these gaps drop the score to the NEEDS_MORE_INFO band.

4.20 CASE20. Principles of Economics to Macroeconomics

Source: ECON 101 Principles of Economics. **Target:** ECON 1010 Macroeconomics. **Decision:** APPROVE_WITH_BRIDGE at score 80, HIGH confidence.

All 6 of 6 required topics (supply and demand, market structures, macroeconomic indicators, fiscal and monetary policy, international trade, economic growth) are covered. All 3 of 3 outcomes (analyze markets, evaluate macro policies, apply economic reasoning to real world problems) are matched. Only credits are unknown. Score caps at 80. HIGH confidence reflects the comprehensive match.

4.21 CASE21. Social Dance to Social Dance

Source: DANCE 150 Social Dance. **Target:** DANC 1240 Social Dance. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

The source covers 5 of 6 required dance styles (waltz, foxtrot, swing, cha cha, partner leading and following). All 3 of 3 outcomes (perform basic steps in multiple styles, demonstrate proper partner technique and timing, apply social dance etiquette) are matched. Credits are unknown. Score caps at 80.

4.22 CASE22. Elementary Statistics to Introductory Statistics with Lab

Source: STAT 200 Elementary Statistics. **Target:** MATH 2100L Introductory Statistics with Lab. **Decision:** NEEDS_MORE_INFO at score 70, MEDIUM confidence.

All 6 of 6 required statistics topics and all 3 of 3 outcomes are matched. The target requires a lab, but the source's lab status is unknown. Credits are also unknown. Both caps apply, dropping the score to 70. The engine asks for more information rather than approving on a strong topic match alone.

4.23 CASE23. Nursing Practice to Pathophysiology

Source: NURS 1010 Nursing Practice. **Target:** NURS 2260 Pathophysiology. **Decision:** DENY at score 0, HIGH confidence.

A nursing practice course teaches patient care skills and safety, not pathology. None of the required pathology topics (cellular injury, inflammation, immune response, organ system pathology) are covered, and none of the pathology outcomes are matched. The two courses share a department prefix but cover different content. Engine correctly denies.

4.24 CASE24. Engineering Intro to Engineering Design

Source: ENGR 1281 Introduction to Engineering. **Target:** ENGR 1020 Introduction to Engineering Design. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

The source covers the engineering design process, teamwork, and communication, matching 3 of 5 required topics. All 3 of 3 outcomes (apply the design process, communicate technical concepts,

describe engineering disciplines) are matched. Engineering ethics and engineering disciplines overview are not explicitly covered. Credits are unknown. Score caps at 80.

4.25 CASE25. Critical Reasoning to Critical Thinking

Source: PHIL 100 Critical Reasoning. **Target:** PHIL 1110 Critical Thinking. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

The source covers logical reasoning, argument analysis, informal fallacies, and deductive and inductive reasoning. All 5 of 5 required outcomes are matched (identify and evaluate arguments, recognize logical fallacies, etc.). Only credits are unknown. Score caps at 80.

4.26 CASE26. Biotech Survey to General Biology

Source: BT 101. **Target:** BIOL 1110 General Biology. **Decision:** NEEDS_MORE_INFO at score 70, MEDIUM confidence.

The source covers cell biology, genetics, and molecular biology (3 of 5 required topics). One outcome (describe the structure and function of cellular components) is matched. Notably, the source includes a lab component that satisfies the lab requirement. However, the topics on evolution and ecology are not covered, the outcome on applying evolutionary principles to biological diversity is missing, and credits are unknown. The combination of content gaps and unknown credits routes this case through NEEDS_MORE_INFO rather than APPROVE_WITH_BRIDGE.

4.27 CASE27. Statistics to Introductory Statistics

Source: STA 2023 Introduction to Statistics. **Target:** MATH 2100 Introductory Statistics. **Decision:** APPROVE_WITH_BRIDGE at score 80, HIGH confidence.

Direct statistics to statistics transfer. The source covers descriptive statistics, probability, and inferential statistics, matching all 6 required topics. All required outcomes (summarize and visualize data, perform hypothesis tests, apply regression analysis) are explicitly matched. Credits are unknown. Score caps at 80. HIGH confidence reflects the complete topic and outcome alignment.

4.28 CASE28. Aviation to Digital Logic and Computer Organization

Source: AA 101 (aeronautics and astronautics). **Target:** CPSC 2310 Digital Logic and Computer Organization. **Decision:** DENY at score 10, HIGH confidence.

Aerospace content does not overlap with computer architecture content. None of the required digital circuits or assembly topics are matched. None of the required outcomes (designing circuits, assembly language, CPU datapath) are matched. The lab component is also not aligned. Engine correctly denies.

4.29 CASE29. Computer Information Systems Survey to Programming I

Source: CIS 105 (computer literacy and productivity applications survey). **Target:** CPSC 1110 Programming I. **Decision:** DENY at score 0, HIGH confidence.

A computer literacy survey course teaches users how to operate productivity software, not how to program. None of the required programming topics or outcomes are covered. The course outcomes

focus on hardware, networking, and business applications, not programming skills. Engine correctly denies.

4.30 CASE30. Chemical Engineering Thermodynamics to Chemical Engineering Thermodynamics

Source: CHE 205. **Target:** ENCH 3030 Chemical Engineering Thermodynamics. **Decision:** APPROVE_WITH_BRIDGE at score 80, MEDIUM confidence.

Direct thermodynamics to thermodynamics transfer. The source covers the first and second laws, PVT behavior of fluids, and equilibria. All 6 of 6 required outcomes (apply thermodynamic laws to engineering processes, analyze phase and chemical equilibria, etc.) are matched. Credits are unknown. Score caps at 80.

5 Configurable Rules Verification

Per spec: Configurable rules: min grade, lab requirement parity, min contact hours, must include topic X, expiration rules (course too old). Bridge plan generator for partial matches.

5.1 Implementation status

All five configurable rules are implemented in the deterministic engine (`decision_engine/contracts.py`) and documented in the LLM system prompt (`prompts/policy.md`). An earlier configuration gap was corrected so that `_format_target_for_prompt` in `llm_decision.py` now passes `min_grade`, `min_contact_hours`, `max_course_age_years`, and `must_include_topics` to the per call LLM prompt under a Configurable Veto Rules (enabled) section, so policy config changes are no longer silently ignored by the LLM path.

Rule	Engine	LLM	30 case	Targeted test
Lab requirement parity	`contracts.py: 409-441`	yes	yes (CASE14, 15, 19, 22)	yes
`min_grade`	`contracts.py: 521-550`	yes	not exercised	yes
`min_contact_hours`	`contracts.py: 553-585`	yes	not exercised	yes
`max_course_age_years`	`contracts.py: 588-618`	yes	not exercised	yes
`must_include_topics`	`contracts.py: 621-641`	yes	not exercised	yes
Bridge plan (partial matches)	`contracts.py: 393-502`	yes	yes (13 of 13)	yes

The four rules not exercised by the 30 case suite are off in `policy.yaml` by default. All transcripts are recent with passing grades, and there are no policy level mandatory topics for these targets. To verify the rules independent of the regression suite, `verify_configurable_rules.py` feeds hand crafted synthetic evidence into `decide()` and asserts the expected gap fires:

```
Configurable Policy Rules. deterministic verification
=================================================================
--- min_grade ---              [PASS] D+ < C threshold fires HARD gap
--- lab parity ---             [PASS] required but missing fires FIXABLE gap
--- min_contact_hours ---      [PASS] 30h < 45h floor fires HARD gap
--- max_course_age_years ---   [PASS] source taken 2016, max 5 years fires HARD gap
--- must_include_topics ---    [PASS] missing 'recursion' fires HARD gap
--- bridge plan generator ---   [PASS] one BridgeItem per unmatched topic or outcome
=================================================================
Passed: 6 of 6
```

5.2 Bridge plan generator coverage

The bridge plan generator emits a `BridgeItem` for each partial match gap:

Gap type	Bridge item template	Type
Credits off by 1	Complete an additional 1 credit bridge component...	course
Credits unknown	Verify the credit hours for ⟨course⟩...	course
Lab required but missing	Take the target lab (or an approved lab equivalent)...	lab
Topic T not matched	Cover the missing topic 'T'...	self_study
Outcome O not matched	Demonstrate the missing learning outcome: 'O'...	project

In this run, every APPROVE_WITH_BRIDGE case (13 of 13) cited at least one structured gap addressable by the bridge generator, and every NEEDS_MORE_INFO case (6 of 6) produced an explicit info request listing the unknown evidence fields.

6 Known Limitations

1. **LLM nondeterminism on score values.** Even with `temperature=0.2`, GPT-4o produces small score variations run to run on borderline cases (typically up to 10 score points within the same decision band) and occasionally flips between adjacent confidence levels (MEDIUM and HIGH). Across multiple runs on identical code observed during this evaluation, all 30 decisions remained stable at the bucket level. Only the underlying score and confidence drifted. The deterministic engine in `contracts.py` serves as a backstop for cases where the LLM produces invalid output.

2. **Sample size.** 30 cases is the project's designated regression suite. A larger labeled corpus would tighten reported confidence intervals on the 100% accuracy figure.

7 Conclusions

- The decision engine correctly classifies all 30 cases. Strict accuracy is 100%, with 100% per class on every decision bucket.
- Confidence is calibrated. Every HIGH confidence and MEDIUM confidence verdict in this run is correct. No LOW confidence outputs were produced, which is consistent with the calibration design (LOW is reserved for unusual ambiguity).

- The security defender works end to end. The CASE07 prompt injection probe is detected and rejected before any decision is made, with score 8 and five distinct trigger patterns matched. Adversarial input cannot reach the LLM.
- Lab parity, credit cap, and topic coverage rules behave correctly when extraction surfaces partial evidence. The system downgrades to APPROVE_WITH_BRIDGE (extraction known fields strong, credits unknown) or NEEDS_MORE_INFO (lab evidence required and absent, or topic gaps with credits unknown) instead of forcing approve or deny.
- All five configurable veto rules (min_grade, lab parity, min_contact_hours, must_include_topics, max_course_age_years) are implemented, passed to the LLM prompt when enabled, and verified by 6 of 6 deterministic unit tests.
- The partial match bridge plan generator (one of the two features called out in the spec) is fully functional. Every APPROVE_WITH_BRIDGE case produces structured gaps that map to bridge items.

The Decision Logic and Policy Engine component meets the spec's requirements for equivalency scoring, gap analysis, configurable rules, bridge plan generation, and calibrated confidence on the 30 case evaluation suite.

A Reproduction

```
Configurable Policy Rules. deterministic verification
====================================================================
--- min_grade ---             [PASS] D+ < C threshold fires HARD gap
--- lab parity ---            [PASS] required but missing fires FIXABLE gap
--- min_contact_hours ---     [PASS] 30h < 45h floor fires HARD gap
--- max_course_age_years ---  [PASS] source taken 2016, max 5 years fires HARD gap
--- must_include_topics ---   [PASS] missing 'recursion' fires HARD gap
--- bridge plan generator ---  [PASS] one BridgeItem per unmatched topic or outcome
====================================================================
Passed: 6 of 6
```

Local environment requirements

- Python 3.11 or later with `pdfplumber`, `pypdf`, `pikepdf`, `pytesseract`, `pdf2image`, `ocrmypdf`, `openai`, `sqlalchemy`, `psycopg2`.
- Tesseract OCR with `eng.traineddata` accessible (env var `TESSDATA_PREFIX` pointing to the directory containing the data file).
- Poppler (Windows: `C:/tools/poppler/poppler-25.12.0/Library/bin/`, or set `POPPLER_PATH` env var).
- Postgres database matching `Database/db_schema.sql`, with `DATABASE_URL` configured.